together by the strong nuclear force: gluon particles (the messenger particles of the strong force) stream between quarks, “gluing” them together. Experiments have shown that these gluons are highly energetic, and since Einstein’s $E=mc^2$ tells us that energy (E) can manifest itself as mass (m), we learn that the gluons inside protons and neutrons contribute a significant fraction of these particles’ total mass. Thus, a more precise picture is to think of the molasseslike drag force of the Higgs ocean as giving mass to *fundamental* particles such as electrons and quarks, but when these particles combine into composite particles like protons, neutrons, and atoms, other (well understood) sources of mass also come into play.

Physicists assume that the degree to which the Higgs ocean resists a particle’s acceleration varies with the particular species of particle. This is essential, because the known species of fundamental particles all have different masses. For example, while protons and neutrons are composed of two species of quarks (called *up-quarks* and *down-quarks*: a proton is made from two ups and a down; a neutron, from two downs and an up), over the years experimenters using atom smashers have discovered four other species of quark particles, whose masses span a wide range, from .0047 to 189 times the mass of a proton. Physicists believe the explanation for the variety of masses is that different kinds of particles interact more or less strongly with the Higgs ocean. If a particle moves smoothly through the Higgs ocean with little or no interaction, there will be little or no drag and the particle will have little or no mass. The photon is a good example. Photons pass completely unhindered through the Higgs ocean and so have no mass at all. If, to the contrary, a particle interacts significantly with the Higgs ocean, it will have a higher mass. The heaviest quark (it’s called the *top-quark*), with a mass that’s about 350,000 times an electron’s, interacts 350,000 times more strongly with the Higgs ocean than does an electron; it has greater difficulty accelerating through the Higgs ocean, and that’s why it has a greater mass. If we liken a particle’s mass to a person’s fame, then the Higgs ocean is like the paparazzi: those who are unknown pass through the swarming photographers with ease, but famous politicians and movie stars have to push much harder to reach their destination.[9]